Khaled Waleed Ba Tais

AI & Computer Vision Engineer · Edge AI

khaled0wleed@gmail.com | Kuala Lumpur, Malaysia | LinkedIn | GitHub | Portfolio

SUMMARY

Third-year Computer Science (AI) student and edge-AI builder who ships real, on-device computer-vision products — not just notebooks. Comfortable across the full pipeline — data labeling, model training with PyTorch/CNNs, and deployment to Raspberry Pi and Jetson with real-time, offline inference. Award-winning (Darb 2026 Winner; Top 33 of 439 teams at the Asian Hackathon for Green Future 2026). Seeking an AI / Computer Vision internship to turn models into deployable products.

TECHNICAL SKILLS

Languages & ML: Python, PyTorch, OpenCV, MediaPipe, scikit-learn (Random Forest), NumPy, Pandas

Computer Vision: object detection, recognition & tracking, image segmentation, real-time video inference

Edge AI & Deployment: NVIDIA Jetson, Raspberry Pi, Hailo (26 TOPS) accelerator, model quantization & optimization, on-device / offline inference

Embedded & IoT: ESP32, Arduino, sensors, MQTT pipelines, 3D printing / CAD enclosures

Tools & Workflow: Git & GitHub, Hugging Face Spaces, Vercel, dataset labeling, model training & evaluation

PROJECTS

ClimaShield — Edge-AI Smart Agriculture

Live · Code

- Built an autonomous rooftop-garden guardian that detects crop and leaf stress from a live camera and makes on-device decisions with a Random Forest model, auto-deploying shade when heat/moisture thresholds are crossed.
- Runs fully offline on Raspberry Pi with real-time inference; integrated sensors and ESP32/MQTT with a companion monitoring dashboard, and built a full hardware "grow-rig" prototype.
- Selected **Top 33 of 439 teams (22 countries)** for the Asian Hackathon for Green Future 2026 (For Green Future Foundation, Vingroup).

Tech: Python, PyTorch, OpenCV, scikit-learn, Raspberry Pi, ESP32, MQTT

AquaGuard — Real-Time Drowning-Detection Vision System

Live · Code

- Developed an AI smart camera that monitors a pool and detects drowning and at-risk situations in real time, firing an instant alert to a phone.
- All detection runs on-device in real time (edge AI, no cloud dependency); validated in a real-world pool test.

Tech: Python, PyTorch, OpenCV, edge deployment

SignBridge — Two-Way Sign-Language Translator

Demo · Live · Code

- Built a real-time translator that converts Arabic sign language into speech and speech back into text, enabling two-way communication between Deaf and hearing users.
- Shipped a public interactive demo on Hugging Face Spaces and a web app on Vercel; **winner of the Darb 2026 hackathon**.

Tech: Python, PyTorch, OpenCV, MediaPipe, Hugging Face

AWARDS & RECOGNITION

- **Darb 2026 — Winner (MYR 2,000)**; honored by the Ambassador of Yemen and the Yemeni Students Union in Malaysia.
- **Asian Hackathon for Green Future 2026 — Top 33 of 439 teams** from 22 countries; on-site finalist in Hanoi (For Green Future Foundation, Vingroup).
- **ENERtech Asia 2026 — TV interview** on AI and using it to drive real-world decision-making.

LEADERSHIP & ACTIVITIES

Team Leader — led multidisciplinary teams to ship products under hackathon deadlines with a clear, goal-driven methodology. Active in industry events including **AI Expo Korea (Seoul)**, Cytron IoT & Mechatronics, and hardware / thermal-imaging expos; regularly attends technical conferences alongside professors and mentors.

EDUCATION

Taylor's University, Malaysia — Third Year · Expected 2027

Bachelor of Computer Science (Artificial Intelligence) — focus on deep learning, computer vision & embedded systems.